Parametric Equations

Section 1: First and Second Derivatives Parametrically

1 A curve is defined by $x(t) = t^2 + 1$ and $y(t) = t^3 - 3t$. Find $\frac{dy}{dx}$ at $t = 2$.

a. $\frac{9}{4}$

b. $\frac{3}{2}$

c. $\frac{4}{9}$

d. 9

2 Which formula gives the second derivative $\frac{d^2y}{dx^2}$ in parametric form?

a. $\frac{\frac{d}{dt}[\frac{dy}{dx}]}{\frac{dx}{dt}}$

b. $\frac{d^2y/dt^2}{d^2x/dt^2}$

c. $\frac{d}{dt}[\frac{dy}{dx}]$

d. $\frac{\frac{dy}{dt}}{\frac{d^2x}{dt^2}}$

3 For $x(t) = t^2$ and $y(t) = t^3$, find $\frac{d^2y}{dx^2}$.

a. $\frac{3}{4t}$

b. $\frac{3}{2}$

c. $\frac{6t}{2}$

d. $\frac{3}{4t^2}$

Section 2: Horizontal and Vertical Tangents

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4 At what value of t does the parametric curve $x = t^3 - 3t, y = 2t^2$ have a horizontal tangent line?

a. $t = 0$

b. $t = 1$

c. $t = -1$

d. $t = \pm 1$

5 At what values of t does the curve above have vertical tangent lines?

a. $t = \pm 1$

b. $t = 0$

c. $t = 3$

d. $t = \pm\sqrt{3}$

6 Find the equation of the tangent line at $t = 1$ to the curve $x(t) = e^{2t}, y(t) = 3t^2 - 1$.

a. $y - 2 = \frac{3}{e^2}(x - e^2)$

b. $y - 2 = 3(x - e^2)$

c. $y = \frac{6}{e^2}x$

d. $y - 1 = \frac{3}{e^2}(x - e^2)$

Section 3: Arc Length of Parametric Curves

7 Which integral represents the arc length L of a parametric curve from $t = a$ to $t = b$?

a. $\int_a^b \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2}\,dt$

b. $\int_a^b (\frac{dx}{dt} + \frac{dy}{dt})dt$

c. $\int_a^b \sqrt{1 + (\frac{dy}{dx})^2}\,dt$

d. $\int_a^b \sqrt{x(t)^2 + y(t)^2}\,dt$

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8 Find the arc length of the curve $x = 3\cos t$, $y = 3\sin t$ from $t = 0$ to $t = \pi$.

a. 3π

b. 6π

c. 9π

d. $\frac{3\pi}{2}$

9 Find the arc length of $x = 2t^{3/2}$, $y = 3t$ from $t = 0$ to $t = 3$.

a. 14

b. 16

c. 18

d. 12

Section 4: Speed and Total Distance Traveled

10 A particle moves in the xy-plane with velocity vector $\mathbf{v}(t) = \langle 4t, 3 \rangle$. What is its speed at $t = 1$?

a. 5

b. 7

c. 4

d. $\sqrt{7}$

11 What is the total distance traveled by the particle with velocity $\mathbf{v}(t) = \langle 2t, 2 \rangle$ from $t = 0$ to $t = 2$?

a. $\int_0^2 \sqrt{4t^2 + 4}\, dt$

b. $\int_0^2 (2t + 2) dt$

c. $\sqrt{16 + 4}$

d. $2\sqrt{5}$

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12 When is a parametric particle at rest in the coordinate plane?

a. When both $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$ simultaneously

b. When $\frac{dy}{dx} = 0$

c. When $\frac{dx}{dt} = 0$

d. When speed is undefined

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